

8255 Programmable Peripheral Interface (PPI)

The **8255** is a **programmable input/output (I/O) device** that works as an **interface between the microprocessor and peripheral devices**. Peripheral devices are hardware devices like keyboard, printer, sensors, etc. The main purpose of the 8255 is to allow the **microprocessor to communicate with these devices using parallel data transfer**.

The 8255 PPI is programmable, which means the user can configure it according to the needs of the system. Depending on the program, it can control how data moves between the microprocessor and external devices. This flexibility makes it very useful in many microprocessor-based systems.

The 8255 contains **24 input/output pins** which are divided into **three 8-bit ports**. These ports are used for transferring data between the processor and external devices. The three ports are **Port A, Port B, and Port C**. Each port has **8 data lines**.

Port C is special because its 8 bits are divided into **two parts of 4 bits each**. The lower four bits are called **PC0–PC3 (Port C lower)** and the higher four bits are called **PC4–PC7 (Port C upper)**. The ports are also organized into two control groups. **Group A** controls **Port A and the upper part of Port C**, while **Group B** controls **Port B and the lower part of Port C**.

Because of this structure, the 8255 can easily manage data transfer between the processor and multiple external devices.

Modes of 8255 The 8255 can work in two main modes.

Types of Modes

1. **Bit Set Reset (BSR) Mode**
2. **I/O Mode**

Bit Set Reset (BSR) Mode

In this mode, the **individual bits of Port C can be set or reset**. This means a specific bit of Port C can be turned **ON (1)** or **OFF (0)** using instructions from the processor. This mode is mainly used for **control operations**.

I/O Mode

In this mode the ports are used for **data input and output operations**. The I/O mode is divided into three types.

Types of I/O Modes

1. **Mode 0 – Simple Input/Output**
2. **Mode 1 – Input/Output with Handshaking**
3. **Mode 2 – Bidirectional I/O with Handshaking**

Mode 0: Simple Input/Output

Mode 0 is the **basic input/output mode** of the 8255. In this mode, each port can be programmed either as an **input port** or an **output port**. It is the simplest way to transfer data between devices.

Working Principle (Mode 0)

- Each port works as either **input or output**.
- **No handshaking signals** are used.
- **No interrupt support**.
- Input data is **buffered**.
- Output data is **latched**.

Mode 1: Input/Output with Handshaking

Mode 1 is used when the **two devices communicate at different speeds**. In such situations, special signals called **handshaking signals** are used to control the data transfer.

Handshaking ensures that the receiving device is ready before data is sent. This prevents data loss.

In this mode, **Port A and Port B can be used as input or output ports**. Some lines of **Port C are used for handshaking signals** to control the communication.

Working Principle (Mode 1)

- Supports **handshaking signals**.
- Used when devices operate at **different speeds**.
- **3 lines of Port C** are used for handshaking for each port.
- **Interrupt logic is supported**.
- Data in ports is **latched**.

Mode 2: Bidirectional I/O with Handshaking

Mode 2 is similar to Mode 1 but with an important difference. In Mode 2, the port can **send and receive data in both directions**. This is called **bidirectional communication**.

Because of this feature, Mode 2 allows the processor and external device to **exchange data both ways** using handshaking signals.

Working Principle (Mode 2)

- Supports **bidirectional data transfer**.
- Uses **handshaking signals**.

- Allows both **input and output operations**.
- Mainly used with **Port A**.

Architecture of 8255-*****

The internal structure of 8255 contains several functional units that control data transfer between the CPU and the ports.

Data Bus Buffer

The **Data Bus Buffer** connects the internal bus of the 8255 with the **system data bus**. It allows communication between the processor and the ports.

When the CPU writes data, the buffer sends the data **from CPU to the ports or control registers**. When the CPU reads data, the buffer transfers data **from ports or status registers to the CPU**.

Read/Write Control Logic

This unit controls the **internal operations of the 8255**. It manages the transfer of **data, control words, and status information**.

When the processor sends an address, this unit identifies the correct port and generates commands for the required operation.

Group A and Group B Control

The 8255 divides the ports into two groups.

- **Group A-Controls Port A,Controls upper bits of Port C**
- **Group B-Controls Port B,Controls lower bits of Port C**

The CPU sends control words to these groups, and they operate the ports according to the command.

CS (Chip Select)

- Enables communication between **8255 and the processor**.
- **Active LOW signal** enables data transfer.

RD (Read)

- Indicates that the CPU is performing a **read operation**.
- Data is sent **from 8255 to CPU**.

WR (Write)

- Indicates that the CPU is performing a **write operation**.

- Data is sent **from CPU to ports or control register**.

A0 and A1

- Used to **select the port** (Port A, B, C or Control Register).
- Connected with the **lower bits of the address bus**.

RESET

- **Active HIGH signal**.
- Clears control registers and sets all ports to **input mode**.

Ports are set to input mode during reset to **protect the circuit from damage**.

40-Pin Configuration of 8255-xxxxxx

The 8255 has **40 pins** in total.

- **24 pins** are used for **I/O ports**.
- **8 pins** are used for **data bus lines**.
- These data lines are **bidirectional tri-state lines** that allow data transfer between the processor and 8255.
- **Pin 26** is **VCC** which provides power.
- **Pin 7** is **ground**.

Mode 2: Bidirectional I/O with Handshaking

Mode 2 is a special operating mode of the **8255 Programmable Peripheral Interface (PPI)**. In this mode, data can move **in both directions** between the processor and the peripheral device. This means the port can **send data to the device and also receive data from it**. Because data moves in two directions, it is called **bidirectional communication**.

In this mode, **Group A ports** are mainly used. **Port A works as a bidirectional data bus**, meaning it can both input and output data. The **upper four bits of Port C (PC7–PC4)** are used for **handshaking signals**. These signals help control the communication and make sure data is transferred correctly between devices.

The **lower four bits of Port C (PC3–PC0)** are not used for handshaking in this mode. Instead, they can be used for **normal input/output operations**.

Since the data bus is bidirectional, the processor only sends data when the **peripheral device requests it**. This helps prevent errors and ensures that data is transferred only when the receiving device is ready. In Mode 2, Port B is not affected by Port A's bidirectional operation. Port B can still operate in Mode 0 or Mode 1. If Port B is used in Mode 1, then the lower bits of Port C are used for handshaking signals.

Because of these features, Mode 2 is very useful when a device needs **two-way communication with the processor**.

Feature	Mode 0	Mode 1	Mode 2
Working Principle	Simple input/output operation. Ports directly send or receive	Input/output operation with handshaking signals to synchronize data transfer	Bidirectional data transfer with handshaking . Data can move both from CPU to device
Direction of	Unidirectional (either input or output).	Unidirectional but synchronized using	Bidirectional communication.
Handshaking	Not used.	Used to control and synchronize data transfer.	Used for bidirectional communication.
Interrupt	Not supported.	Supported.	Supported.
Ports Used	Port A, Port B, and Port C can be used as	Port A and Port B can operate with handshaking; some bits of	Only Port A works as bidirectional data bus; Port C
Complexity	Very simple.	Moderate complexity.	Most complex among the three
Typical Usage	Simple devices like LEDs, switches, or	Devices requiring controlled data transfer such as printers or	Devices requiring two-way communication such as data

- **Mode 0 → Simple I/O**
- **Mode 1 → I/O with Handshaking**
- **Mode 2 → Bidirectional I/O with Handshaking**

In data communication, **synchronizing means coordinating the data transfer between two devices so that data is sent only when the receiving device is ready**.

Mode	Synchronization	Data Direction
Mode 0	No synchronization	Simple input or output
Mode 1	Synchronization using handshaking	Unidirectional (either input or output)
Mode 2	Synchronization using handshaking	Bidirectional (both input and output)

Control Word Register (CWR)

The **Control Word Register** is used to control the operation of the **8255 ports**. The value stored in this register is called the **control word**. This control word determines how each port will work, such as whether the port will be **input or output**, and which **mode of operation** will be used.

The control word is written into the control register by the processor. To access this register, the **address lines A1 and A0 must both be HIGH**. This register is **write-only**, which means the processor can **write data into it but cannot read data from it**.

The **most important bit in the control word is D7**. This bit decides whether the 8255 will operate in **I/O mode or Bit Set Reset (BSR) mode**.

- If **D7 = 1**, the 8255 works in **I/O mode**.
- If **D7 = 0**, the 8255 works in **BSR mode**.

When the 8255 works in **BSR mode**, only the bits of **Port C** are affected. The operation of **Port A** and **Port B** remains unchanged.

Steps to Communicate with Peripheral Devices

To communicate with peripheral devices using the 8255, the processor must follow some steps.

1. Determine the **addresses of Port A, Port B, Port C, and the control register** using the chip select logic and address lines **A1 and A0**.
2. Write the **control word into the control word register**.
3. Use **input and output instructions** to transfer data through Ports A, B, or C.

Assigning Input and Output Ports in 8255

The control word contains **8 bits (D0–D7)**. Each bit has a specific function that decides how the ports will operate.

Bit D0

- Controls **Port C Lower (PC3–PC0)**
- **1 → Input port, 0 → Output port**

Bit D1

- Controls **Port B**
- **1 → Input port, 0 → Output port**

Bit D2

- Selects **mode of Port B**
- **0 → Mode 0, 1 → Mode 1**

Bit D3

- Controls **Port C Upper (PC7–PC4)**
- **1 → Input port, 0 → Output port**

Bit D4

- Controls **Port A**
- **1 → Input port, 0 → Output port**

Bits D5 and D6

- Used to select the mode of Port A. These bits decide whether Port A works in **Mode 0, Mode 1, or Mode 2**

Bit D7

- Selects **operating mode**
- **1** → I/O Mode, **0** → BSR Mode

Question:

Use 8255 in **Mode 0** where

- **Port A = Output**
- **Port B = Input**
- **Port C = Output**

Step-by-step solution

Mode = **I/O Mode** → D7 = 1

Mode 0 for Port A → D6 = 0, D5 = 0

Port A Output → D4 = 0

Port C Upper Output → D3 = 0

Port B Input → D1 = 1

Port B Mode 0 → D2 = 0

Port C Lower Output → D0 = 0

So the control word becomes:

1 0 0 0 0 0 1 0

Binary: **10000010** Hexadecimal: **82H**